What makes a fungal species? Investigating speciation concepts in the polymorphic tiger sawgill *Letninus tigrinus.*

Alexander J. Bradshaw[1], Thomas Rhoel[1], Humza Musharraf[1], Carlos Perez-Gazca[1], Isaac Perkis[1], Javier F. Tabima[1], David S. Hibbett[1]

Clark University[1]

Introduction

Species delimitation in fungi is challenging because morphology is often plastic, convergent evolution is common, and gene flow may occur across large geographic distances. Genomic datasets now allow species boundaries to be evaluated using multiple lines of evidence. Here we investigate species limits in the tiger sawgill, *Lentinus tigrinus* (Bull.) Fr., a saprotrophic wood-decay fungus found along riverbanks across North America and Eurasia.

Lentinus tigrinus is morphologically polymorphic, producing both agaricoid forms with exposed gills and secotioid forms with enclosed spore-bearing structures, and previous studies have suggested that it represents a complex of cryptic lineages unresolved by standard DNA barcoding.

Here we anlayze 64 genomes across the genus *Lentinus*, including populations of *L. tigrinus* from North America and Eurasia including historical type specimens from fungarium collections. Using phylogenomic, population genomic analyses, and hybridization experiments, we evaluated multiple species recognition criteria. These analyses consistently support the recognition of two species: *Lentinus tigrinus* (Bull.) Fr. in Eurasia and *Lentinus squamulosus* (Morgan) Bradshaw, Roehl, Tabima, and Hibbett comb. nov. in North America. This study demonstrates how integrating historic specimens, genomic data, and reproductive evidence can resolve longstanding taxonomic questions in fungi while also providing information on the natural history and evolution of cryptic taxa.

Methodology

Phylogenomic analysis

1,550 single copy BUSCO[1] genes conserved across Polyporales were used for phylogenomic analysis. Tree topology was determined using weighted-hybrid ASTRAL[2] analysis of individual ML gene trees, branch lengths were then estimated using IQ-TREE 3 with automatic model finder[3,4].

Population Genomics

Variants were called against a newly created hybrid long-read and short-read reference assembly of *Lentinus squamulosus* with the Genome Analysis Toolkit (GATK)[5] for 10 specimens, 5 from North America and 5 from Eurasia. Post filtering for quality and only informative sites with no missing data yielded 4,351,007 SNPs for analysis. Population structure was assessed using both a maximum-likelihood clustering approach with ADMIXTURE[6] and multivariate clustering (Discriminant Analysis of Principal Components; DAPC) using vcfR [7].

Hybridization assays

To assess hybrid viability, monokaryon strains were generated from single-spore isolates (SSIs) derived from wild sporophores. Four parental dikaryons were constructed: one North American, two Eurasian, and one hybrid (North America × Europe). Dikaryons were cultured on malt yeast extract agar, transferred to sawdust–bran fruiting jars, and fruited under controlled conditions (12 h light/dark, 93% humidity, ≤850 ppm CO_2). Spores from mature sporophores were plated to generate SSIs, which were screened for absence of clamp connections to confirm monokaryotic mycelium. Growth rates were measured for 12 North American, 12 hybrid, and 12 Eurasian SSIs.

Works cited

[1]Simão, F. A., Waterhouse, R. M., Ioannidis, P., Kriventseva, E. V. & Zdobnov, E. M. BUSCO: assessing genome assembly and annotation completeness with single-copy orthologs. *Bioinformatics* **31**, 3210–3212 (2015).

[2]Zhang, C., Nielsen, R. & Mirarab, S. ASTER: A Package for Large-Scale Phylogenomic Reconstructions. *Molecular Biology and Evolution* **42**, msaf172 (2025).

[3]Wong, T. *et al.* IQ-TREE 3: Phylogenomic Inference Software using Complex Evolutionary Models. Preprint at https://doi.org/10.32942/X2P62N (2025).

[4]Kalyaanamoorthy S, Minh BQ, Wong TKF, Haeseler A von, Jermiin LS. 2017. ModelFinder: fast model selection for accurate phylogenetic estimates. Nature Methods 14:587–589.

[5]Auwera, G. A. V. de & O'Connor, B. D. *Genomics in the Cloud: Using Docker, GATK, and WDL in Terra*. (O'Reilly, Beijing Boston Farnham Sebastopol Tokyo, 2020).

[6]Zhou, H., Alexander, D. & Lange, K. A quasi-Newton acceleration for high-dimensional optimization algorithms. *Stat Comput* **21**, 261–273 (2011).

[7]Knaus, B. J. & Grünwald, N. J. VCFR: a package to manipulate and visualize variant call format data in R. *Molecular Ecology Resources* **17**, 44–53 (2017).

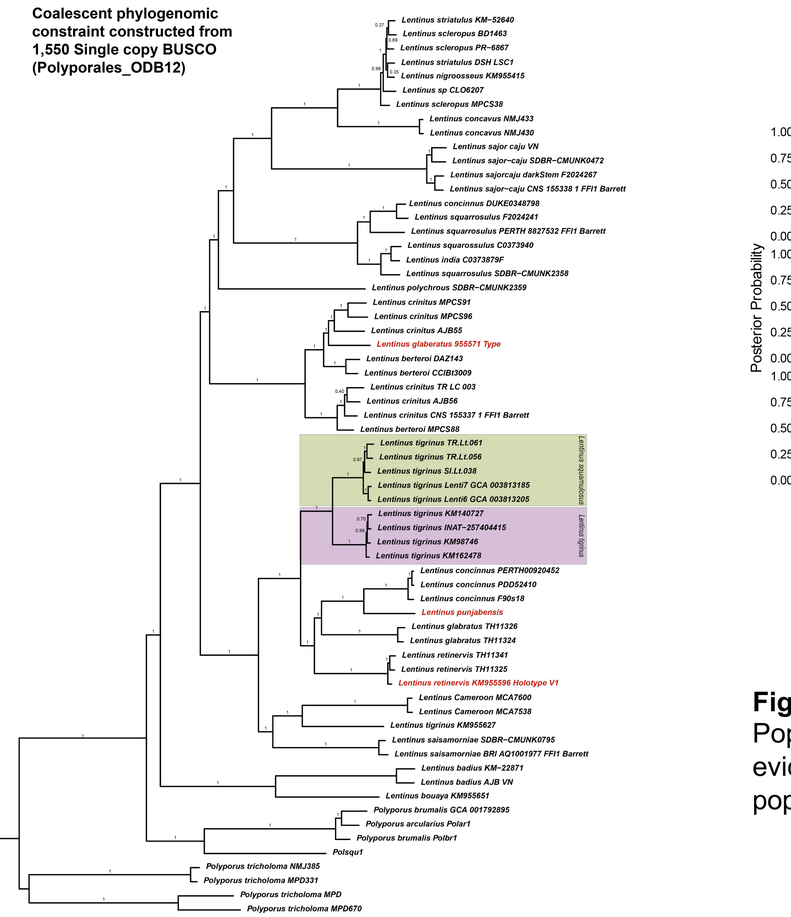


Figure 1: Phylogenomic reconstruction
Lentinus tigrinus and Lentinus squamulosus form monophyletic clades with full bootstrap support.

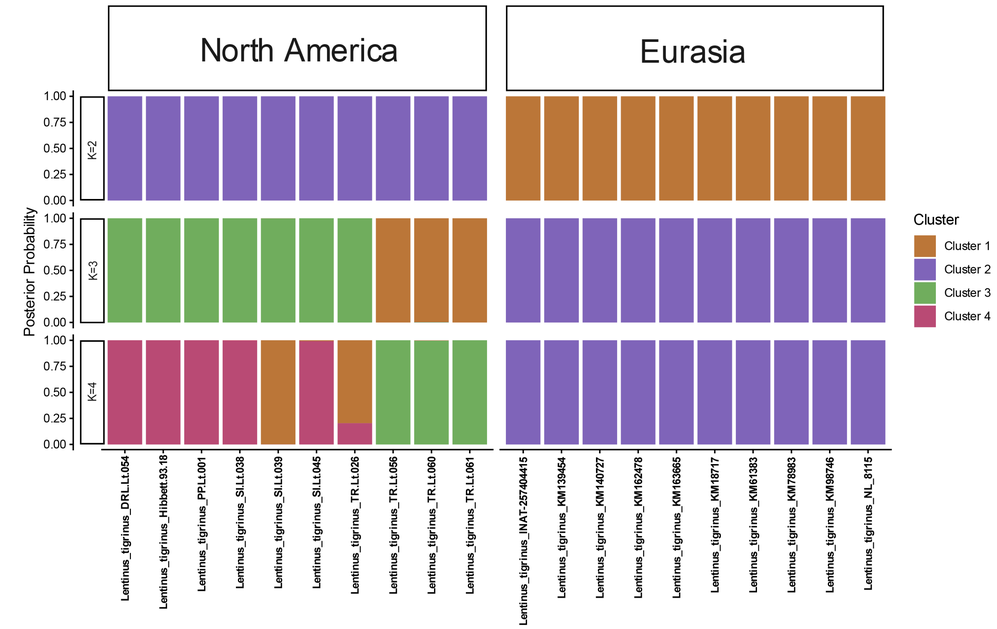


Figure 2: Population ADMIXTURE
Population structure supports two theoretical populations (k=2), with no evidence of gene flow occurring between North American and Eurasian populations.

Lentinus tigrinus represents two distinct lineages from North America and Eurasia.

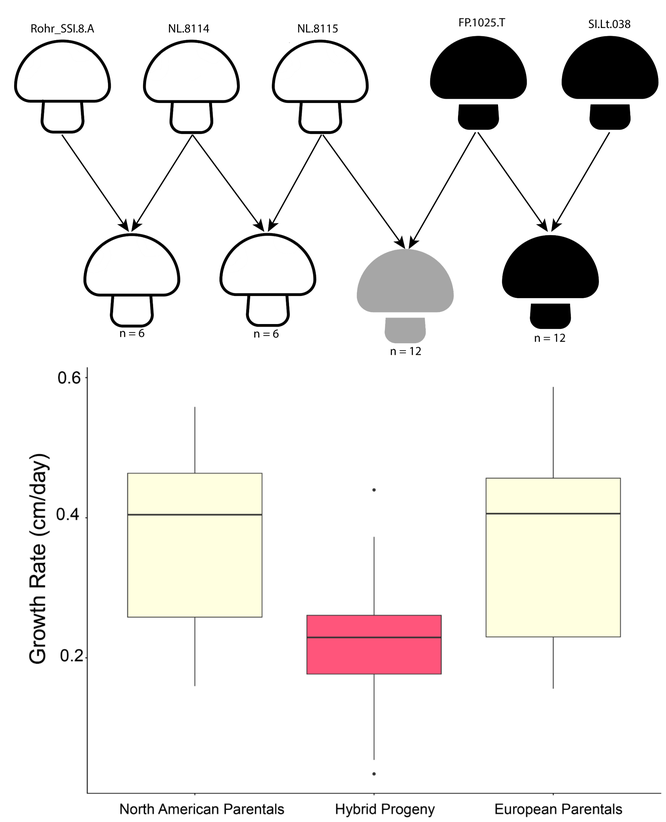
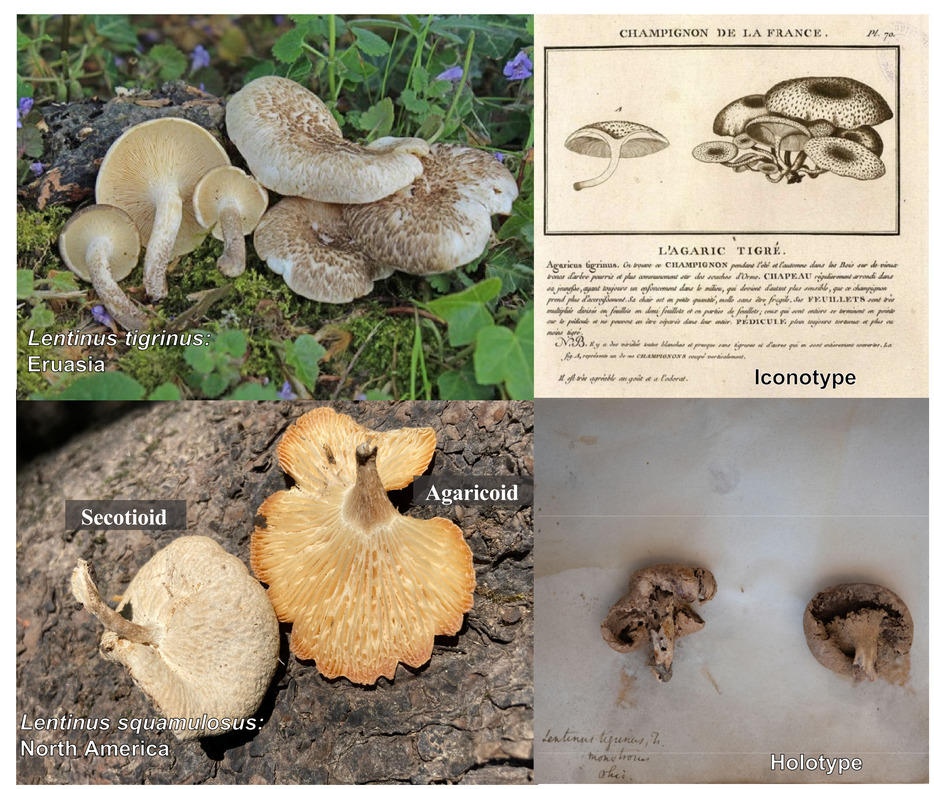


Figure 3: Hybridization growth rates
Hybrid progeny show reduced growth rates and vigor compared to North American and European lineages.



Funding and Support

Award #:2333266

SCAN ME